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Elements de la Théorie d'information
Problem Set 9

Problem 1 Maximum entropy

I'll try to do a) with mathematically accurate notation
and for b) I'll use more flexible notation similar to the course.

minimize/maximize constraint(s)

Lagrange multiplier (General case): $\mathcal{L}(p, \lambda) = f(p) + \sum_i \lambda_i (g_i(p) - a_i)$

Definition of a functional: $J: \begin{array}{l} \mathcal{F} \to \mathbb{R} \\ y(x) \mapsto \int_{x_1}^{x_2} f(x, y, y', y'', \dots, y^{(n)})\, dx \end{array}$

with $\mathcal{F} = \{ y(x) \mid x: \mathbb{R} \to \mathbb{R} \}$ the function space

Here we have $f(p(x)) = -\int_a^b p(x) \ln p(x)\, dx = h(X)$

$$g(p(x)) = \int_a^b p(x)\, dx = 1 \ (= a)$$

and want to maximize the functional $J := J[p(x)] \equiv -\int_a^b p(x) \ln p(x)\, dx + \lambda \left(\int_a^b p(x)\, dx - 1 \right)$

It's an optimization problem, J must verify the Euler Lagrange equation to be stationary

$$\frac{\delta J}{\delta p(x)} = \frac{\partial f}{\partial p} - \frac{d}{dx}\left(\frac{\partial f}{\partial p'} \right) = 0$$

$$= \left(-p(x) \cdot \frac{1}{p(x)} - 1 \cdot \ln p(x) + \lambda \cdot 1 \right) - \frac{d}{dx}(0) = 0$$

$$= -1 - \ln p(x) + \lambda = 0$$

$$\Rightarrow \ln p(x) = \lambda - 1$$

$$p(x) = e^{\lambda - 1}$$

now we search for λ using:

$$\frac{\delta J}{\delta \lambda} = 0 \Leftrightarrow \int_a^b p(x)\, dx = 1 \Rightarrow \int_a^b e^{\lambda-1}\, dx = e^{\lambda-1} \Big|_a^b = e^{\lambda-1}(b-a) = 1$$

$$\Rightarrow e^{\lambda-1} = \frac{1}{b-a} = p(x) \quad \left(h(X) = \int_{x_0} \frac{1}{b-a} \log(b-a) = \log(b- \right.$$

Finally with $x \in [a, b]$ the uniform distribution has the maximal differential entropy

b) we have

$$\mathcal{L}(p(x), \lambda_1, \lambda_2) = -\int_0^\infty p(x) \ln p(x)\, dx + \lambda_1\left(\int_0^\infty p(x)dx - 1\right) + \lambda_2\left(\int_0^\infty x\, p(x)dx - \mu\right)$$

$$\frac{\partial \mathcal{L}}{\partial p(x)} = 0 = -p(x)\cdot\frac{1}{p(x)} - 1\cdot \ln p(x) + \lambda_1 + \lambda_2 x = -\ln p(x) - 1 + \lambda_1 + \lambda_2 x$$

$$\Rightarrow \ln p(x) = \lambda_1 + \lambda_2 x - 1 \quad \Rightarrow \quad p(x) = e^{\lambda_1 + \lambda_2 x - 1}$$

Looking for λ's:

$$\frac{\partial \mathcal{L}}{\partial \lambda_1} = 0 \Leftrightarrow \int_0^\infty e^{\lambda_0 + \lambda_2 x - 1} dx = 1 \Leftrightarrow \int_0^\infty e^{\lambda_2 x} = e^{-\lambda_0 + 1}$$

$$\Rightarrow \frac{1}{\lambda_2} e^{\lambda_2 x}\Big|_0^\infty \overset{\lambda_2 < 0}{=} -\frac{1}{\lambda_2} = e^{-\lambda_1 + 1}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_2} = 0 \Leftrightarrow \int_0^\infty x e^{\lambda_1 + \lambda_2 x - 1} dx = \mu \Leftrightarrow \int_0^\infty x e^{\lambda_2 x} = \mu e^{-\lambda_1 + 1}$$

$$\Rightarrow \frac{e^{\lambda_2 x}(\lambda_2 x - 1)}{\lambda_2^2}\Big|_0^\infty \overset{\lambda_2 < 0}{=} \frac{1}{\lambda_2^2} = \mu e^{-\lambda_1 + 1}$$

$$\begin{cases} -\frac{1}{\lambda_2} = e^{-\lambda_1 + 1} \\ \frac{1}{\lambda_2^2} = \mu e^{-\lambda_1 + 1} \end{cases} \Rightarrow \begin{cases} -\lambda_1 + 1 = \ln\left(-\lambda_2^{-1}\right) \\ \frac{\lambda_2}{\lambda_2^2} = -\mu \end{cases} \Rightarrow \begin{cases} \lambda_1 = \ln(\mu^{-1}) + 1 \\ \lambda_2 = -\mu^{-1} \end{cases}$$

$$\Rightarrow p(x) = e^{\lambda_1 + \lambda_2 x - 1} = e^{\ln(\mu^{-1}) - \frac{x}{\mu} + 1 - 1} = \frac{1}{\mu} e^{-\frac{x}{\mu}}$$

Finally with $x \in [0, \infty)$ (and $E[x] = \mu$), the exponential distribution has the maximal differential entropy.

$$\left(h(x) = -\int_{\mathcal{X}} \frac{1}{\mu}\cdot e^{-\frac{x}{\mu}} \ln\left(\frac{1}{\mu}\cdot e^{-\frac{x}{\mu}}\right) dx = \ln(\mu) - \frac{1}{\mu}\cdot\int_{\mathcal{X}} -\frac{x}{\mu} e^{-\frac{x}{\mu}} dx\right)$$

$$= \log(\mu) + 1$$

Problem 2

$X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$

$X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$

$X_1 \perp X_2$

$$Y = X_1 + X_2 \Rightarrow \begin{cases} \mu_Y = \mu_1 + \mu_2 \\ \sigma_Y^2 = \sigma_1^2 + \sigma_2^2 \; (+2\underbrace{Cov(X_1, X_2)}_{\substack{=0 \\ X_1 \perp X_2}}) \end{cases}$$

$$f_Y(x) = \frac{1}{\sqrt{2\pi}\,\sigma_Y} e^{-\frac{1}{2}\left(\frac{x-\mu_Y}{\sigma_Y}\right)^2}$$

a) $$h(Y) = -\int_{-\infty}^{\infty} f_Y(x) \ln f_Y(x)\,dx = -\int_{-\infty}^{\infty} f_Y(x) \ln\left(\frac{1}{\sqrt{2\pi}\,\sigma_Y} e^{-\frac{1}{2}\left(\frac{x-\mu_Y}{\sigma_Y}\right)^2}\right) dx$$

$$= -\int_{-\infty}^{\infty} f_Y(x)\left(-\ln(\sqrt{2\pi}\,\sigma_Y) - \frac{(x-\mu_Y)^2}{2\sigma_Y^2}\right) dx = \ln(\sqrt{2\pi}\,\sigma_Y)\underbrace{\int_{-\infty}^{\infty} f_Y(x)\,dx}_{1} + \int_{-\infty}^{\infty} f_Y(x)\frac{(x-\mu_Y)^2}{2\sigma_Y^2}\,dx$$

$$= \ln(\sqrt{2\pi}\,\sigma_Y) + \frac{1}{2\sigma_Y^2}\int_{-\infty}^{\infty} f_Y(x)(x-\mu_Y)^2\,dx$$

$$= \ln(\sqrt{2\pi}\,\sigma_Y) + \frac{1}{2\sigma_Y^2}\sigma_Y^2,$$

$$= \frac{1}{2}\ln(2\pi\sigma_Y^2) + \frac{1}{2}\ln e$$

$$= \frac{1}{2}\ln\left(2\pi e(\sigma_{x_1}^2 + \sigma_{x_2}^2)\right)$$

b) $$f_Y(y) = f_{X_1+X_2}(y) = \int_{-\infty}^{\infty} f_{X_1X_2}(x, y-x)\,dx$$

$$\overset{X_1 \perp X_2}{=} \int_{-\infty}^{\infty} f_{X_1}(x)\cdot f_{X_2}(y-x)\,dx = (f_{X_1} * f_{X_2})(y)$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\,\sigma_1} e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2} \frac{1}{\sqrt{2\pi}\,\sigma_2} e^{-\frac{1}{2}\left(\frac{y-x-\mu_2}{\sigma_2}\right)^2} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\,\sigma_Y \sqrt{2\pi}\,\frac{\sigma_1\sigma_2}{\sigma_Y}} e^{\left(-\frac{\sigma_2^2(x-\mu_1)^2 + \sigma_1^2(y-x-\mu_2)^2}{2\sigma_1^2\sigma_2^2}\right)} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\,\sigma_Y}\frac{1}{\sqrt{2\pi}\,\frac{\sigma_1\sigma_2}{\sigma_Y}} e^{\left(-\frac{x^2 - \frac{2\sigma_1^2(y-\mu_2)+\sigma_2^2\mu_1}{\sigma_Y^2} + \frac{\sigma_1^2(y^2+\mu_2^2-2y\mu_2)+\sigma_2^2\mu_1^2}{\sigma_Y^2}}{2\left(\frac{\sigma_1\sigma_2}{\sigma_Y}\right)^2}\right)} dx \quad (*)$$

$$= \frac{1}{\sqrt{2\pi}\,\sigma_Y} e^{\left(-\frac{(y-\mu_Y)^2}{\sigma_Y^2}\right)} \cdot \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\,\frac{\sigma_1\sigma_2}{\sigma_Y}} e^{\left(-\frac{\left(x - \frac{\sigma_1^2(y-\mu_2)+\sigma_2^2\mu_1}{\sigma_Y^2}\right)^2}{2\left(\frac{\sigma_1\sigma_2}{\sigma_Y}\right)^2}\right)} dx$$

$$= \frac{1}{\sqrt{2\pi}\,\sigma_Y} e^{\left(-\frac{(y-(\mu_1+\mu_2))^2}{2\sigma_Y^2}\right)}$$

$$(*) = \frac{\left(-\left(x - \frac{\sigma_1^2(y-\mu_2)+\sigma_2^2\mu_1}{\sigma_Y^2}\right)^2 - \frac{(\sigma_1^2(y-\mu_2)+\sigma_2^2\mu_1)^2}{\sigma_Y^2} + \frac{\sigma_1^2(y-\mu_2)^2+\sigma_2^2+\mu_1^2}{\sigma_Y^2}\right)}{2\left(\frac{\sigma_1\sigma_2}{\sigma_Y}\right)^2}$$

$$h(Y) = \frac{1}{2}\ln(2\pi e\sigma_Y^2).$$

Problem 3

$$h(aX) = h(Y) = -\int f_Y(y) \log f(y)\, dy \quad ; \quad f_Y(y) = \frac{1}{|a|} f_X(y/a)$$

$$= -\int \frac{1}{|a|} f_X(y/a) \log\left(\frac{1}{|a|} f_X(y/a)\right) dy$$

$$= -\int \frac{1}{|a|} f_X(y/a) \log(f_X(y/a))\, dy + \int \frac{1}{|a|} f_X(y/a) \log(|a|)\, dy$$

Changing var $z = y/a$ $dz = \frac{dy}{a}$

$$= -\int f_X(z) \log(f_X(z))\, dz + \log|a| \cdot \underbrace{\int f(y)\, dy}_{1}$$

$$h(aX) = h(X) + \log|a|$$

Problem 4

$Y = X_1 + X_2$

$E[X_1^2] = \sigma_1^2 = E[(X_1-\mu_1)^2] \Rightarrow \mu_1 = 0$

$E[X_2^2] = \sigma_2^2 = E[(X_2-\mu_2)^2] \Rightarrow \mu_2 = 0$

a) $X_1 \perp X_2 \Rightarrow Var[Y] = Var[X_1+X_2] = Var[X_1] + Var[X_2]$ ($\cancel{2}Cov(X_1, X_2)$ $= 0$)

for fixed variance maximum entropy is with normal pdf $\Rightarrow Y \sim N(\underbrace{\mu_1+\mu_2}_{0}, \underbrace{\sigma_1^2+\sigma_2^2}_{\sigma_Y^2})$

$$h(Y) = \frac{1}{2}\log_2\left(2\pi e\left(\sigma_1^2+\sigma_2^2\right)\right)$$ See problem 2

b) $X_1 \not\perp X_2 \Rightarrow Var[Y] = Var[X_1] + Var[X_2] + \underbrace{2Cov(X_1, X_2)}_{\sigma_{X_1X_2}}$

$$\Rightarrow h(Y) = \frac{1}{2}\log_2\left(2\pi e\left(\sigma_Y^2\right)\right) = \frac{1}{2}\log_2\left(2\pi e\left(\sigma_1^2+\sigma_2^2 + \underbrace{Cov(X_1, X_2)}_{\sigma_{X_1X_2}}\right)\right)$$